

SKILL WEEK-1(OSSP)

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TASK 2:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;

    printf("Current Process PID : %d\n", getpid());
    printf("Current Process PPID : %d\n", getppid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        printf("Child PPID: %d\n", getppid());

        printf("\nChild will execute 'ls' using exec()\n");
        sleep(2);

        execl("/bin/ls", "ls", "-l", NULL);
        perror("exec failed");
        exit(1);
    }
    else
    {
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        printf("\nParent waiting for child...\n");
        wait(NULL);

        printf("\nChild process has terminated.\n");
        printf("Parent process completed.\n");
    }

    return 0;
}
```

OUTPUT:

```

[varshithcomple@VARSHITHs-MacBook-Air ~ % gcc fork.c -ofork
clang: warning: overriding deployment version from '16.0' to '26.0' [-Woverriding-deployment-version]
[varshithcomple@VARSHITHs-MacBook-Air ~ % ./fork
Current Process PID : 50923
Current Process PPID : 50465
Parent PID : 50923
Child PID : 50925

Parent waiting for child...
Child PID : 50925
Child PPID: 50923

Child will execute 'ls' using exec()
total 576
drwx-----@ 7 varshithcomple staff 224 1 Jun 14:30 Applications
drwx-----@ 6 varshithcomple staff 192 19 Aug 09:35 Desktop
drwx-----@ 8 varshithcomple staff 256 17 Aug 09:09 Documents
drwx-----@ 24 varshithcomple staff 768 20 Aug 13:38 Downloads
-rwxr-xr-x 1 varshithcomple staff 33720 19 Aug 07:34 exec
-rw-r--r-- 1 varshithcomple staff 757 19 Aug 07:34 exec.c
-rwxr-xr-x 1 varshithcomple staff 33768 25 Jul 09:28 first
-rw-r--r-- 1 varshithcomple staff 1896 25 Jul 09:11 first.c
-rwxr-xr-x 1 varshithcomple staff 33800 20 Aug 13:42 fork
-rw-r--r-- 1 varshithcomple staff 934 20 Aug 13:42 fork.c
drwx-----@ 99 varshithcomple staff 3168 23 Jul 12:13 Library
drwx----- 5 varshithcomple staff 160 5 Jun 08:55 Movies
-rwxr-xr-x 1 varshithcomple staff 33648 19 Aug 07:47 ms
-rw-r--r-- 1 varshithcomple staff 1044 19 Aug 07:47 ms.c
drwx-----+ 5 varshithcomple staff 160 18 Feb 2026 Music
-rwxr-xr-x 1 varshithcomple staff 33544 19 Aug 07:27 notes
-rw-r--r-- 1 varshithcomple staff 455 19 Aug 07:28 notes.c
-rw-r--r-- 1 varshithcomple staff 497 19 Aug 07:18 notes.txt
-rwxr-xr-x 1 varshithcomple staff 33792 19 Aug 07:52 op
-rw-r--r-- 1 varshithcomple staff 1819 19 Aug 07:52 op.c
drwx-----+ 6 varshithcomple staff 192 1 Jun 14:32 Pictures
drwxr-xr-x+ 5 varshithcomple staff 160 18 Feb 2026 Public
-rwxr-xr-x 1 varshithcomple staff 33992 11 Aug 12:34 task1
-rw-r--r-- 1 varshithcomple staff 1468 11 Aug 12:27 task1.c
-rw----- 1 varshithcomple staff 1675 11 Aug 12:26 task1.c.save

Child process has terminated.
Parent process completed.
varshithcomple@VARSHITHs-MacBook-Air ~ % █

```

TASK 3:

```
#include <stdio.h>
#include <string.h>

int main()
{
    char input[100];

    while (1)
    {
        printf("myshell> ");

        if (fgets(input, sizeof(input), stdin) == NULL)
        {
            printf("\nInput error or EOF detected.\n");
            break;
        }

        input[strcspn(input, "\n")] = '\0';

        if (strlen(input) == 0)
        {
            continue;
        }

        if (strcmp(input, "exit") == 0)
        {
            printf("Exiting shell...\n");
            break;
        }

        if (strcmp(input, "hello") == 0)
        {
            printf("Hello! Welcome to the interactive shell.\n");
        }
        else if (strcmp(input, "help") == 0)
        {
            printf("Available commands:\n");
            printf("hello - Display greeting\n");
            printf("help - Display help\n");
            printf("exit - Exit the shell\n");
        }
        else
        {
            printf("Unknown command: %s\n", input);
        }
    }

    printf("Program terminated.\n");

    return 0;
}
```

OUTPUT:

```
myshell> hello
Hello! Welcome to the interactive shell.
myshell> elp
Unknown command: elp
myshell> help
Available commands:
hello - Display greeting
help - Display help
exit - Exit the shell
myshell> █
```

TASK 4:

```

#include <stdio.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

#define BUFFER_SIZE 100

int main()
{
    struct termios oldt, newt;
    char buffer[BUFFER_SIZE];
    int index = 0;
    char ch;

    tcgetattr(STDIN_FILENO, &oldt);
    newt = oldt;
    newt.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    printf("Type a command and press Enter.\n");
    printf("Type 'exit' to quit.\n\n");
    printf("> ");

    while (1)
    {
        read(STDIN_FILENO, &ch, 1);

        if (ch == '\n' || ch == '\r')
        {
            buffer[index] = '\0';

            printf("\nYou entered: %s\n", buffer);

            if (strcmp(buffer, "exit") == 0)
            {
                break;
            }

            if (strcmp(buffer, "hello") == 0)
            {
                printf("Hello! Welcome.\n");
            }
            else if (strcmp(buffer, "help") == 0)
            {
                printf("Commands: hello, help, exit\n");
            }
            else if (index == 0)
            {
                printf("No input entered.\n");
            }
            else
            {
                printf("Command received successfully.\n");
            }

            index = 0;
            printf("> ");
        }
        else if (ch == 127 || ch == 8)
        {
            if (index > 0)
            {
                index--;
                printf("\b \b");
                fflush(stdout);
            }
        }
        else if (ch >= 32 && ch <= 126)
        {
            if (index < BUFFER_SIZE - 1)

```

```

                buffer[index++] = ch;
                printf("%c", ch);
                fflush(stdout);
            }
        }

        tcsetattr(STDIN_FILENO, TCSANOW, &oldt);
        printf("\nProgram terminated.\n");

        return 0;
    }
}

```

OUTPUT:

```
[varshithcomple@VARSHITHs-MacBook-Air ~ % nano ms.c
[varshithcomple@VARSHITHs-MacBook-Air ~ % gcc ms.c -oms
clang: warning: overriding deployment version from '16.0' to '26.0' [-Woverriding-deployment-version]
[varshithcomple@VARSHITHs-MacBook-Air ~ % ./ms
myshell> hello
Hello! Welcome to the interactive shell.
myshell> help
Available commands:
hello - Display greeting
help - Display help
exit - Exit the shell
myshell> nano ms.c
Unknown command: nano ms.c
myshell> exit
Exiting shell...
Program terminated.
```